MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University,
New Delhi -110067
Programming in C (CS-401) Mid-I

Total Marks:20 **Date: 07.10.2024** **Time: 2hr**

All Compulsory

1. Write note on:
    a. High level and low level languages, with examples
    b. Translator, Compiler and Assembler in Computer
    c. function for Input and output data, alongwith relevant libraries
    d. Data types
    e. Keywords 2*5=10
2. Write a program to interchange the value of two variables. 3
3. Write a C program for : 'Accepts marks in five subject and calculate the total percentage marks'. 3
4. Discuss the flow chart and C program to print the smaller and larger number of two numbers. 4

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Mid Term Examination-Academic Year 2026-27

Program	:	**MCA**	**Semester**	:	**I**
Course Title	:	**Programming in C**	**Course Code**	:	**CS-401**
Faculty Name	:	**Subhash Chandra Patel**	**Date/Time**	:	**05-10-2026/2:00PM-3:00 PM**

Attempt any four (04), each question carry equal marks.

1. **(A).** What is an algorithm? Write any two characteristics of a good algorithm. **2.5**

   (B). Differentiate between the = and == operators **2.5**

2. **(A).** What is the purpose of the if-else statement? Give suitable example to demonstrate your answer. **2.5**

   (B). Differentiate between while and do-while loops with suitable examples. **2.5**

3. Determine the basic steps involved in program design and development. Illustrate the steps using a simple example. **05**

4. Design an algorithm and write a C program to repeatedly accept numbers from the user and calculate their sum. The program should stop when the user enters 0. **05**

5. Write a C program to input a number and determine whether it is: **05**
   - Positive
   - Negative
   - Zero

6. Write a C program to accept the marks of three subjects and calculate 05
   - Total marks
   - Average marks
   - Percentage

**********Good Luck**********